

Can Breast MRI Help Predict Treatment Response?

Forecasting pretraining made the vessel map matter, but timing complicates the story

Spencer Venancio

with Sarit Bose and Aakrithi Ram · Advisor: Anna Woodard, PhD

Neoadjuvant therapy is given before surgery, and can shrink a tumor.

THE TREATMENT

Before surgery

Neoadjuvant chemotherapy can shrink a tumor and is associated with better outcomes.

THE OUTCOME

≈ 1 in 3

scans came from patients with no remaining invasive cancer at surgery.

THE QUESTION

Can MRI help?

How much signal is there in vessel networks and contrast flow?

The long-term goal is better evidence for tailoring treatment.

Tumors build a messier blood supply

New vessels don't look or behave like normal ones.

Normal vessels: organized, efficient.

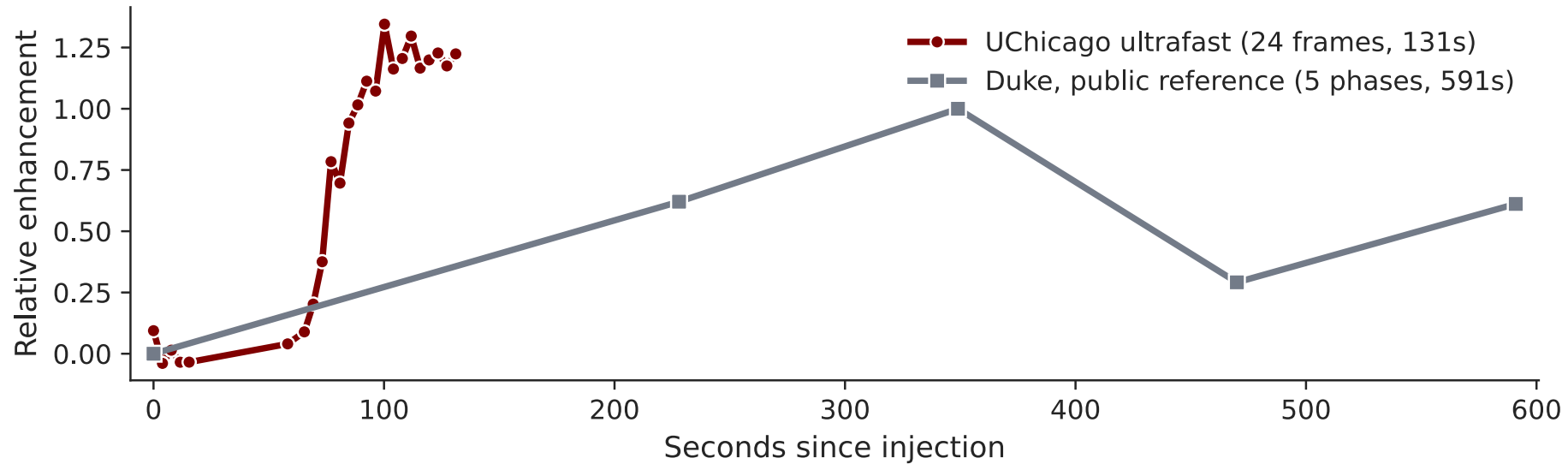
Blood flows through a predictable, well-structured network.

Tumor vessels: dense, tortuous, leaky.

Tumors take a lot of blood to keep growing.
Structure may reflect how aggressive a tumor is.

If the blood vessel structure aligns with tumor behavior, it might help predict response.
That's the biological reason to expect a vessel-based signal.

Ultrafast MRI movie shows more than anatomy



179

labeled exams, 141 patients

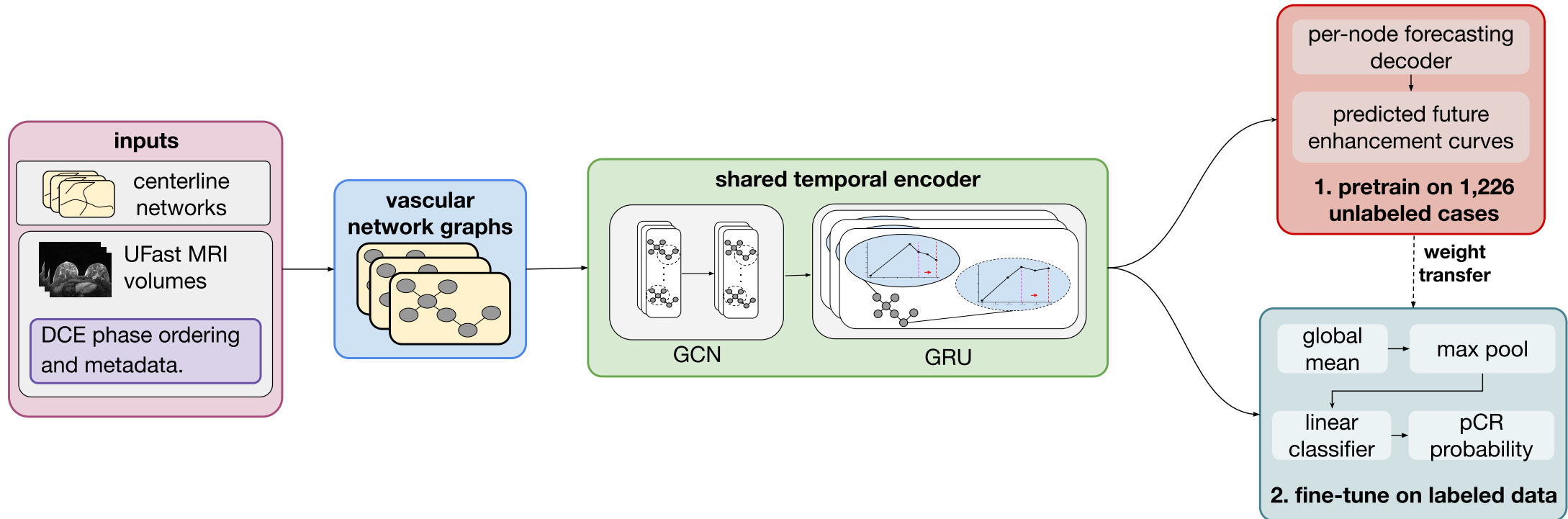
1,226

unlabeled exams, pretraining only

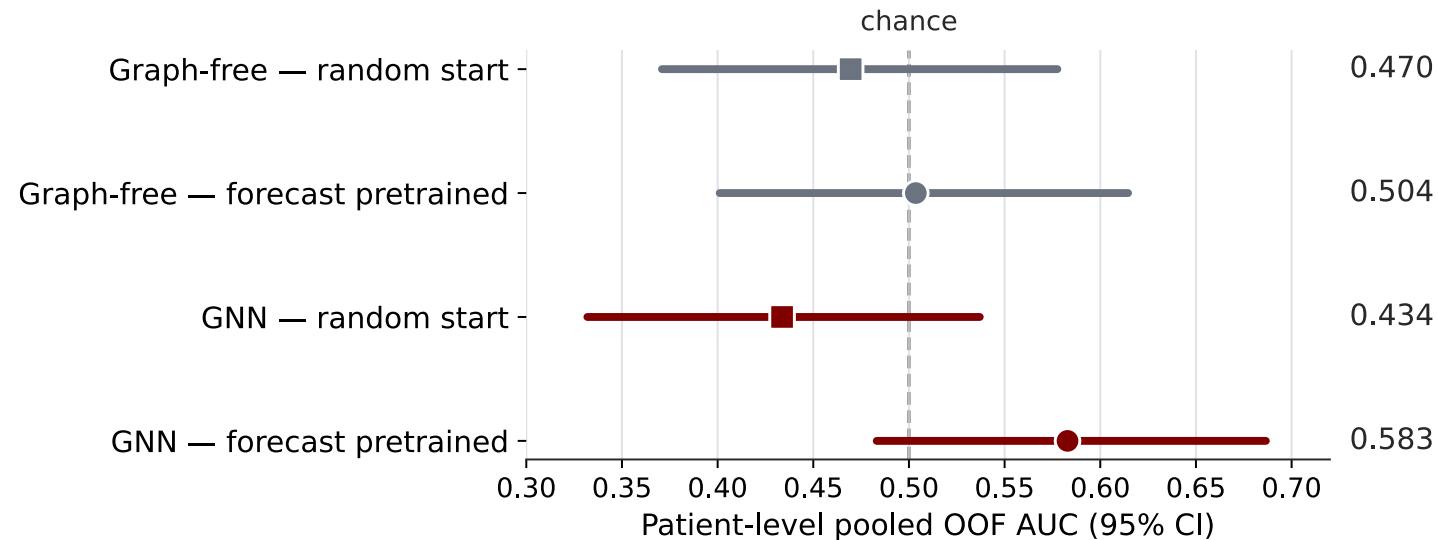
13–24

frames per ultrafast movie

Does the graph structure help? How about pretraining?



Pretraining gave the graph model a real boost



The boost was real: **+0.149 [+0.046, +0.255]** AUC for the GNN, a gain graph-free didn't reliably show. Pretrained, GNN then beat graph-free by **+0.079 [+0.010, +0.155]**.

But the clock may be part of the story

Timing alone scores nearly as well.

A model using only frame count, scan duration, and cadence (no imaging at all) reaches 0.587 AUC, almost identical to the pretrained graph model's 0.583.

The two track each other.

Their predictions correlate (Pearson $r = 0.50$), so acquisition timing is associated with outcome in this cohort.

This raises concern about protocol dependence, but doesn't tell us how much of the signal is technical versus biological. Adjustment could also remove real patient differences.

What should you remember?

1

Forecast pretraining gave the **graph model a real boost**, which then beat the matched simple-summary model.

2

Vessel enhancement still carries only **modest, uncertain** signal on its own.

3

Acquisition timing tracks outcome too. Next, we **separate protocol from biology**.

Teaching the model to anticipate the future made the vessel map matter, but timing still complicates the story.